



Attempt to Measure the Unmeasurable

ucal — the Universe Calendar, in two pages

The irritation

Recombination occurred about 380,000 years after the Big Bang. A good sentence, and its units are a problem. A year is the time Earth takes to circle the Sun — definitionally, not approximately. So an event 13.8 billion years old is described in units defined by a planet that would not exist for another nine billion.

The number is correct. It carries a passenger.

Not a complaint about cosmology: Julian years are stable, agreed and unambiguous, which is what a unit should be. The complaint is that **provenance leaks into arithmetic**. Compute in Earth years and Earth's orbital period sits inside every intermediate value.

What was built

ucal is six Rust crates. Time is an unsigned integer count of Planck-time units — **ticks**, 5.391×10^{-44} s — since a stipulated datum.

Ticks group into tiers, the powers 5^{5k} , each exactly five base-5 digits. The reference rung is the **beat**, 5^{60} ticks ≈ 46.762 ms, which the specification calls the universe second: human-noticeable, with no Earth content. Base five because $5^5 = 3125$ is five base-5 digits — that is the whole reason.

A timestamp is the tick count written in base 5 and grouped in fives, so truncation **is** rounding, prefix comparison **is** chronological comparison, and writing every digit pinpoints one tick.

The cost is that nothing on the ladder is near anything you know. One second is 21.385061835 beats, and the two share a common measure only at the tick.

Three engineering commitments

Unsigned. Nothing precedes the datum. A result that would be earlier is **UCAL-E0020** — an error, not a negative number. The refusal is the answer.

No floating point, anywhere. Not in a signature, a field, an intermediate, or the rendering path; a lint fails the build on any float token. Cosmology is therefore certified interval quadrature over exact rationals — orders of magnitude slower, and it returns an **enclosure** with a proof rather than an estimate with a guess.

One declared boundary. Earth enters through a single exact constant, **SECOND** = $18\,548\,584\,399\,861 \times 10^{30}$ ticks. Conversion in is multiplication and never rounds; conversion out is division and is the only place a rounding mode is chosen. The dependency is not eliminated — that is not available to anyone — it is **localised** to somewhere you can point at and argue about.

The problem with zero

A count needs a zero, and the obvious one is not available.

The published age is 13.787 ± 0.020 Gyr. That uncertainty is 58 digits of ticks, 0.145% of the whole span. Define the datum **as** the measured age and every timestamp inherits the wobble, making the exact arithmetic theatre.

So the datum is **stipulated**: declared, not discovered. And the physical claim about it is kept — cited, with its exact magnitude — in a type that has **no arithmetic operations at all**. Three tests exist whose job is to **fail to build** if anyone ever computes with it.

The philosophical background

The line between what a measurement **establishes** and what it merely **points at** is old, and it does not stay where you put it. Kant was bluntest: the illusion that erodes it is natural, unavoidable, and **persists after diagnosis** — the astronomer knows the moon is no larger at the horizon and goes on seeing it so.

The usual remedy is vigilance, which lasts as long as attention does. In 1922 Florensky — who had the distinction available in his own tradition, fifteen centuries deep — computed a radius from the Lorentz factor outside its domain of validity and identified the resulting surface with the Emyrean. One move, and nothing in his method forbade it.

Kant's First Antinomy supplies the reason the origin cannot be measured at all: both **the world has a beginning and it does not** are demonstrable, and both fail, because both assume a completed totality about which the question has an answer waiting. Stipulating is not a retreat from a better method. There is no better method.

Cusanus, 1440: no proportion holds between finite and infinite, so precision does not constitute approach.

The theological background

Basil of Caesarea, fourth century: the beginning of a road is not the road. A beginning is not a member of the series it begins — which is the datum's content, with no arithmetic.

Gregory of Nyssa makes **διάστημα**, interval, the mark of creat- edness: everything created has it, and what lies beyond has none. Every quantity in this system is an interval, so the title stops being a paradox and becomes a statement about categories.

The Hebrew calendar's epoch is **molad tohu**, “the new moon of chaos” — computational, placed before the event it anchors, and named for its own emptiness. This project spent four chapters explaining that its epoch is stipulated; that one puts it in the name. Its **helek** was chosen for divisibility exactly as **SECOND** was.

And a validity window — assert a parameter where it was measured, warn beyond — is al-Ghazālī's ‘**āda** with the theology removed.

None of this argues that any tradition is correct. The decisions have older, better-named precedents reached for other purposes; reporting that is an argument that the decisions are not eccentric, and nothing more.

What came out of it

Give the mechanism nothing but Earth's rotation and orbit and its first answer is **1/4** — the Julian calendar, with no knowledge of Rome. **97/400** never appears: the Gregorian rule is not a convergent at any depth, and **31/128** is **124× more accurate** with a denominator three times smaller.

The specification claimed twice, in print, that the mechanism reproduces both. It reproduces one. That correction is published, and a build-time assertion now stops it being reversed.

Two more fell out. The Metonic cycle — 235 months in 19 years, known to Babylon — appears from two numbers unaided. And Mars has **no month**: neither moon is one, and the mechanism returns nothing rather than inventing one, because **month-like** is an Earth predicate.

What is claimed, and what is not

A measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing — and that declaration can be enforced mechanically rather than left to the author's discipline.

The first clause is Cusanus'. The second is the contribution: a comment addresses a reader who is paying attention, a runtime check addresses a program that reaches a line, and a **type** addresses everyone who ever writes against the library — including someone who disagrees. They are not persuaded. They are refused, in under a second.

Not claimed: that time began; that the Big Bang happened at tick 0; that the tick is time's smallest unit; that base 5 means anything; that any tradition is right; or that the system is **useful** — no task you have needs a Planck-tick count, and that is the second of three facts rather than an apology.

Where **next.** **cargo install ucal,** or **cargo add ucal-core.** The normative specification is **spec/UCAL-1.1.md**; the twenty-four rules and what enforces each are in **spec/RULES.md**. A 29-page account is **UCAL_INTRO**. The full argument — nine traditions, six samples, and a chapter on what they failed to establish — is **Life, the Universe, and God**, 251 pages.